Terrain Classification for Small Legged Robots from Tactile Data

L G Vivyn Krishnaa · PES1UG24CS248
Likhith B · PES1UG24CS249

github.com/cockroachyie/Terrain-Classification-for-Small-Legged-Robots

01 · PROBLEM

What is the robot walking on?

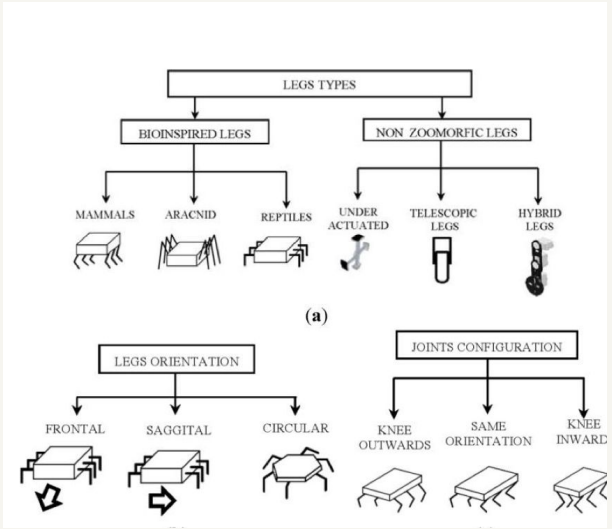
8
individual terrains

4
terrain classes

Small legged robots move very differently on hard floors, slippery tiles, grass or sand. If the robot knows the terrain, it can change its gait to save energy or avoid slipping.

SAIL-R, a small hexapod from Stanford, has no camera - but its C-shaped front legs carry a capacitive tactile array

Following Choi & Thomasson, we predict the terrain from the tactile signal of a single step, at two levels: the 8 individual terrains and 4 terrain classes (high-friction HF, low-friction LF, deformable D, granular G).



Types of Hexapod Robots

02 · DATASET

SAIL-R tactile dataset: 5,990 steps

414

Raw signal

6 channels × 69 samples at 217 Hz - 5 normal-force taxels + 1 shear taxel

33

Hand-designed features

Summed normal force (peak, area, mean), per-taxel peaks, ratios & means, shear min/max/mean, motor RPM & current at peak

3

Gait control inputs

Motor RPM, Ts/Tc ratio, ωslow - part of the 33, and they cannot be recovered from the tactile signal

Stratified 70 / 15 / 15 split · train-only normalisation · 3 random splits (seeds 42, 7, 2026)

02 · LABELS

Balanced terrains, unbalanced classes

Share of steps per class (≈750 steps per terrain)



High friction (HF)
Asphalt

Low friction (LF)
Lab floor, laminate wood

Deformable (D)
Cork, grass, gravel, pebble

Granular (G)
Sand

03 · APPROACH

A tuned baseline, a CNN and a fusion

BASELINE

SVM, Pearson VII kernel

PUK kernel on the 33 standardised features - same setup as earlier SAIL-R papers, but with C and σ tuned on the validation set.

LEARNED

1D CNN

Input 6 × 69 → four Conv1d blocks (24 channels) → dropout 0.5 → two fully-connected layers (hidden 128).

COMBINED

Feature fusion

Hand-made features are concatenated with the learned CNN features before the FC layers.

Three CNN variants - each trained twice: once for the 4 classes, once for the 8 terrains

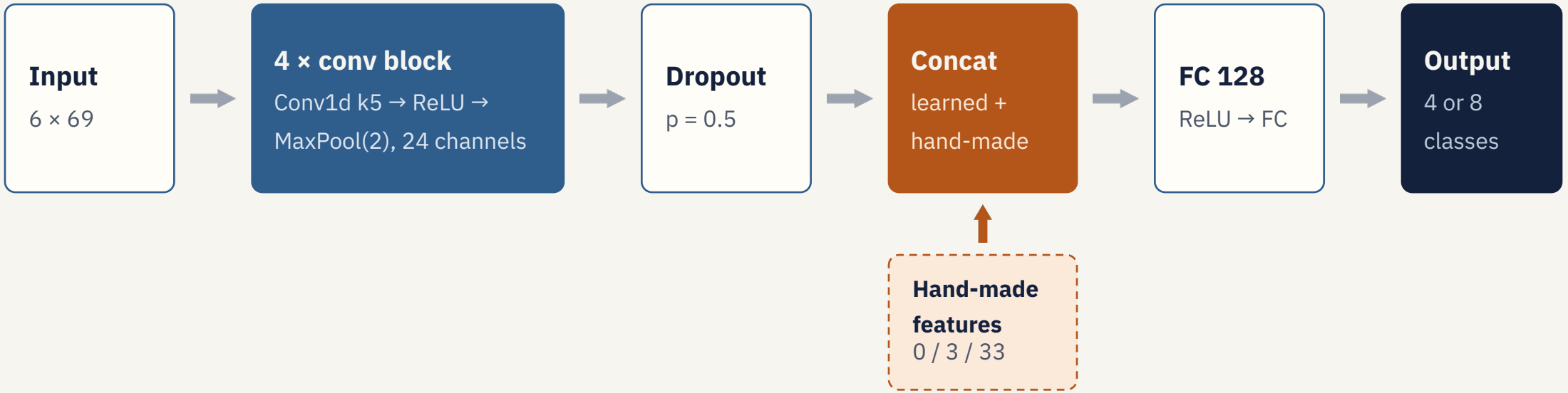
Raw Data · no extra features

Raw + Control Inputs · 3 gait parameters

Raw + All Features · all 33

03 · MODEL

The 1D CNN with feature fusion



Cross-entropy · Adam, lr 10^{-3} · batch 64 · up to 200 epochs · best-validation epoch kept (patience 50)

Repositories

File	What it does
data.py	Loads the CSV, reshapes 414 values into 6 × 69, splits and normalises
models.py	The CNN and the precomputed-kernel PUK SVM
train.py	Training loop and metrics
scripts/run_all.py	Runs all 8 experiments; saves results, confusion matrices, curves, checkpoints
scripts/demo.py	Classifies held-out steps; --stream walks the robot across all 8 terrains

STACK

Python · PyTorch · scikit-learn

~8 min

per seed for all 8 experiments, on a laptop CPU

05 · RESULTS

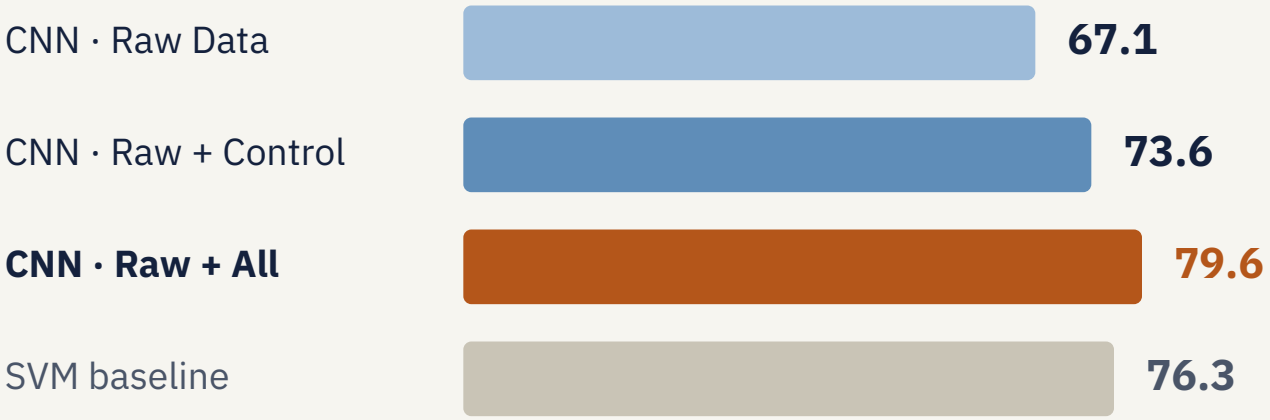
Raw + All Features is best on both tasks

Model	Class (4)	Terrain (8)	Paper class	Paper terrain
SVM baseline (PUK, 33 features)	91.9 ± 0.4	76.3 ± 1.7	84	70
CNN · Raw Data	86.4 ± 0.5	67.1 ± 0.9	87	69
CNN · Raw + Control Inputs	90.5 ± 1.5	73.6 ± 1.5	93	76
CNN · Raw + All Features (best)	92.6 ± 1.2	79.6 ± 0.6	97	80

05 · FEATURE FUSION

Combining hand-made features

Terrain (8-way) test accuracy, %



Bars on a 0–100 scale

67 → 80%

terrain accuracy after adding all 33 hand-made features

+6 / +4

points (terrain / class) from just the 3 gait control inputs

The same terrain looks different at a different gait - and the CNN cannot learn the gait from the tactile signal alone.

05 · ERRORS

HF vs LF is the hardest split

True ↓	HF	LF	D	G
HF	**0.77**	**0.22**	0.02	0.00
LF	0.06	**0.92**	0.02	0.00
D	0.00	0.01	**0.98**	0.00
G	0.00	0.00	0.03	**0.97**

Predicted → · Raw + All Features, 4 classes, seed 42

22% of HF steps read as LF

Asphalt (μ ≈ 1.0) gets confused with lab floor and laminate (μ ≈ 0.5) — and HF has half as many samples as LF.

Deformable and granular are easy

0.98 and 0.97 on the class matrix; at terrain level, sand reaches 98% and grass 87%.

05 · TAKEAWAYS

How we compare to the paper

REPRODUCED

CNN numbers close to the paper’s

Raw: 86 / 67 vs 87 / 69

Raw + All: 93 / 80 vs 97 / 80

class / terrain, ours vs paper

STRONGER BASELINE

Tuned SVM: 92 / 76

vs the paper’s 84 / 70 — and it beats the raw-only CNN on both tasks.

BEST OVERALL

Raw + All Features

Best on class and terrain, with the lowest variance on terrain (± 0.6).

06 · CONCLUSIONS

~93% class, ~80% terrain

The CNN beats a well-tuned SVM only when it also gets the hand-made features - and the gait control inputs matter most, because they cannot be inferred from touch.

- Random split over steps: steps from the same run can land in both train and test (no run IDs in the dataset)
- The model needs one complete step before it can predict

- Run-wise split
- Higher-rate shear sensing to separate HF from LF
- One model predicting class and terrain together
- Sequence models (GRU / Transformer) for mid-step prediction

REFERENCES

Thank you

H. Choi, R. Thomasson, *Terrain Classification for Small Legged Robots Using Deep Learning on Tactile Data*, Stanford CS229.

X. A. Wu et al., *Tactile Sensing and Terrain-Based Gait Control for Small Legged Robots*, IEEE T-RO, 2019.

CODE

`github.com/cockroachyie/Terrain-Classification-for-Small-Legged-Robots`